

INDU 211 · Rapid Review Sheet (1-Page Cheatsheet)

Part 3: Chapter 4 — Facilities Location, Layout & Material Handling

1. Strategic Levels & Location Drivers

- **3 Decision Levels:** 1. **General Location** (country/region) | 2. **Exact Site** (zoning, parcel, soil) | 3. **Internal Layout** (machines, aisles, utilities).
- **Time Horizon:** Strategic (3–10+ yrs); high sunk capital; requires IEs, accountants, legal, marketing, executives.
- **Proximity Rules:**
 - **Market-Oriented:** Bulky, fragile, perishable goods, or rapid delivery (potato chips, bakeries, beer).
 - **Raw Material-Oriented:** Weight-losing, bulky raw materials (steel mills, paper mills, ore smelters).
- **The 30% to 95% Rule: Material handling = 30% to 95% of manufacturing cost** (variable & quantifiable → prime IE focus).

2. Distance Metrics & Location Formulas

- **Euclidean Distance (L_2):** Straight-line / "as the crow flies": $d_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ (intercity, pipelines, air).
- **Rectilinear Distance (L_1):** Orthogonal grid travel: $d_R = |x_1 - x_2| + |y_1 - y_2|$ (urban streets, factory aisles, AGVs).
 - **Elongation Penalty:** $d_R \geq d_E$. Maximum ratio at 45°: $d_R/d_E = \sqrt{2} \approx 1.414$ (up to 41.4% extra travel).
- **Center of Gravity (Centroid):** Minimizes sum of squared Euclidean distances:
 - **Unweighted:** $\bar{x} = \frac{\sum x_i}{n}$, $\bar{y} = \frac{\sum y_i}{n}$
 - **Weighted:** $\bar{x} = \frac{\sum Q_i x_i}{\sum Q_i}$, $\bar{y} = \frac{\sum Q_i y_i}{\sum Q_i}$ (High-volume destinations pull centroid toward them).
- **Rectilinear 1-Median Rule:** Minimizes weighted rectilinear distance $\sum Q_i (|x - x_i| + |y - y_i|)$.
 - Independently find x^* and y^* at the median coordinate where cumulative weight $\sum Q_i \geq 0.50 \cdot Q_{\text{total}}$.
- **Transportation Model (Least-Cost Method):**
 1. Find cell with lowest unit cost c_{ij} ;
 2. Allocate $\min(\text{Supply}_i, \text{Demand}_j)$;
 3. Reduce row/col;
 4. Repeat.

3. Master 5-Layout Configurations Comparison

Layout Type	Operational Regime	Flow Pattern	WIP Inventory	Unit Handling Cost	Failure Sensitivity	Industrial Examples
Product (Line / Continuous)	High Volume, Low Variety	Progressive straight/U line	Very Low (smooth flow)	Low (fixed conveyors)	Catastrophic (1 machine halts line)	Paper mills, soft drinks, auto assembly lines.
Process (Job Shop / Functional)	Low Volume, High Variety	Intermittent / criss-crossing	Extremely High (queues)	High (variable forklifts)	Low (reroute to same bay)	Machine shops, hospitals, commercial banks.
Cellular (Group Technology)	Medium Vol, Medium Variety	U-shaped cell flow	Low to Moderate	Low within cell	Isolated to single cell	Furniture (chair cells), auto gear/brake cells.
Fixed-Position (Project)	Unit / Project, Heavy/Bulky	Stationary (resources come)	Project-based	High (mobile tools/rigs)	Schedule milestone delays	Shipbuilding, aircraft (Boeing 777), rockets.
Mixed / Hybrid	Multi-echelon manufacturing	Staged combination	Optimized per stage	Balanced	Isolated by buffer zones	Auto (Process → Cellular → Product).

4. Layout Drivers & Industrial Work Environment

- **Drivers for Relayout:** Bottlenecks, high handling cost, safety hazards/accidents, product design changes, volume/mix shifts, morale.
- **IE Focus:** Simultaneous design of layout + material handling; utilities (power drops, compressed air, water); lighting & safety aisles.